

Comparative Statics: System, Jacobian, Determinants, and Sign Test

ECON 320 – Introduction to Mathematical Economics

Fall 2025

System and Base Point

Consider the nonlinear system with parameters (α, β, γ) :

$$\begin{cases} F_1 : x^2 + y + z = \alpha, \\ F_2 : xy - z = \beta, \\ F_3 : y^2 + z^2 = \gamma. \end{cases}$$

At $(x, y, z) = (1, 1, 0)$ the system holds for $(\alpha, \beta, \gamma) = (2, 1, 1)$. We study local effects around this point.

Step 1. Jacobian with respect to (x, y, z)

$$J(x, y, z) = \begin{bmatrix} \partial_x F_1 & \partial_y F_1 & \partial_z F_1 \\ \partial_x F_2 & \partial_y F_2 & \partial_z F_2 \\ \partial_x F_3 & \partial_y F_3 & \partial_z F_3 \end{bmatrix} = \begin{bmatrix} 2x & 1 & 1 \\ y & x & -1 \\ 0 & 2y & 2z \end{bmatrix}.$$

At $(1, 1, 0)$,

$$J = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 2 & 0 \end{bmatrix}.$$

Step 2. Determinant of the Jacobian and IFT

Compute $\Delta = \det J$ at $(1, 1, 0)$:

$$\Delta = 2 \det \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} - 1 \det \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix} + 0 \cdot (\dots) = 2(2) - (-2) = 6.$$

Since $\Delta = 6 \neq 0$, the Implicit Function Theorem applies, so (x, y, z) are differentiable functions of (α, β, γ) locally.

Step 3. Linearization and Cramer's Rule

Total differentiation yields

$$J \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix} = \begin{bmatrix} d\alpha \\ d\beta \\ d\gamma \end{bmatrix}.$$

For a single parameter change, set the right side to the corresponding unit vector and use Cramer's rule:

$$\frac{\partial x}{\partial \theta} = \frac{\Delta_1^{(\theta)}}{\Delta}, \quad \frac{\partial y}{\partial \theta} = \frac{\Delta_2^{(\theta)}}{\Delta}, \quad \frac{\partial z}{\partial \theta} = \frac{\Delta_3^{(\theta)}}{\Delta}, \quad \theta \in \{\alpha, \beta, \gamma\}.$$

With respect to α

Right-hand side $b_\alpha = (1, 0, 0)^\top$.

$$\Delta_1^{(\alpha)} = \det \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & 0 \end{bmatrix} = 2, \quad \Delta_2^{(\alpha)} = \det \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} = 0, \quad \Delta_3^{(\alpha)} = \det \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} = 2.$$

Therefore

$$\boxed{\frac{\partial x}{\partial \alpha} = \frac{1}{3}, \quad \frac{\partial y}{\partial \alpha} = 0, \quad \frac{\partial z}{\partial \alpha} = \frac{1}{3}}.$$

With respect to β

Right-hand side $b_\beta = (0, 1, 0)^\top$.

$$\Delta_1^{(\beta)} = \det \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 2 & 0 \end{bmatrix} = 2, \quad \Delta_2^{(\beta)} = \det \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} = 0, \quad \Delta_3^{(\beta)} = \det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 2 & 0 \end{bmatrix} = -4.$$

Therefore

$$\boxed{\frac{\partial x}{\partial \beta} = \frac{1}{3}, \quad \frac{\partial y}{\partial \beta} = 0, \quad \frac{\partial z}{\partial \beta} = -\frac{2}{3}}.$$

With respect to γ

Right-hand side $b_\gamma = (0, 0, 1)^\top$.

$$\Delta_1^{(\gamma)} = \det \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 0 \end{bmatrix} = -2, \quad \Delta_2^{(\gamma)} = \det \begin{bmatrix} 2 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = 3, \quad \Delta_3^{(\gamma)} = \det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} = 1.$$

Therefore

$$\boxed{\frac{\partial x}{\partial \gamma} = -\frac{1}{3}, \quad \frac{\partial y}{\partial \gamma} = \frac{1}{2}, \quad \frac{\partial z}{\partial \gamma} = \frac{1}{6}}.$$

Step 4. Sign Test Summary

Since $\Delta > 0$, each sign is determined by the numerator's sign:

$$\alpha : (\partial x/\partial\alpha, \partial y/\partial\alpha, \partial z/\partial\alpha) = (+, 0, +),$$

$$\beta : (\partial x/\partial\beta, \partial y/\partial\beta, \partial z/\partial\beta) = (+, 0, -),$$

$$\gamma : (\partial x/\partial\gamma, \partial y/\partial\gamma, \partial z/\partial\gamma) = (-, +, +).$$